

Electronictization-A Foundation for Grid Modernization

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Abstract: Smart grid is the flag under which the US DoE has been mobilizing efforts to modernize the grid. Electronictization is the first step towards a smart modern grid. It is a process that transforms the grid from electrical and electromechanical (EE) to electronic, electrical and electromechanical (EEE), laying down the very basic foundation for the modern grid. All things grid connected (ATGC) has five groups of essential hardware: 1) Grid interface (smart) inverters; 2) Hardware for flexible AC transmissions; 3) Intelligent electronic power transformers (grid scale); 4) Solid-state circuit breaker, current limiters, smart fuses and sensors; and 5) Multi-port bidirectional power & control units. Development and deployment of ATGC will be a grassroots drive to transform the grid from an old passive technology to a new active technology based on electronic power transmission, distribution, processing and protection. Grid modernization represents a win-win-win situation for the environment (Government), consumers, and grid owners/operators.

Keywords: Grid modernization, electronictization, fractal grid structure, energy system, resiliency, fault management, fault isolation, renewable energy integration, all things grid connected.

1 Introduction

The electrical grid is an amazing engineering achievement. In fact, at the turn of the century, the US National Academy of Engineering named “Electrification” as number one among the 20 top engineering achievements in the 20th century for its “vast networks of electricity to provide power for the developed world”^[1]. Over the last century, the grid has evolved into a system of systems. The technology is mature and electricity reliability (availability) is high, while system reliability (power availability) is typically 99.99% or better.

The basic technology of the electrical power grid was developed about a century ago and has been incrementally improved and its operation evolved^[2] as new technologies, such as computing and networking, became available. However, its passive nature has not changed. While it typically works very well, its potential failure modes are equally spectacular: total blackout in a large area, such as the August 2003 US-Canada Blackout and the September 2003 Italy Blackout. Both happened quickly in a matter of 1.5 minutes after the original failure occurred^[3]. Each impacted tens of millions of people, resulting in billions of dollars of economic losses. Grid’s vulnerability to large blackouts is due to an insufficient capability to be fault tolerant and the lack of the ability for fault isolation.

The grid technology developed at the turn of the last century suffers from the following deficiencies: 1) Lack of bidirectional power flow; 2) Lack of fault tolerance; 3) Lack of fault isolation; 4) Slow electro-

mechanical protection (Particularly at the grid edge); 5) Lack of over voltage protection; 6) Lack of over-current limiter; 7) Lack of power limiter; 8) Lack of single point failure prevention mechanism; 9) Over-loaded transmission capacity; and 10) Inefficient and expensive central generation’s operation and maintenance.

To address the energy challenges in the coming century, the US Department of Energy started the smart grid initiative under the Energy Independence and Security Act in 2007. Since then, the smart grid today has evolved to include the application of advanced communications and control technologies and practices to improve reliability, efficiency, and security. Specific technical areas include: 1) Advanced metering infrastructure, which comprises smart meters, communication networks, and information management systems; 2) Customer-based technologies, such as programmable communicating thermostats for residential customers and building energy management systems for commercial and industrial customers; and 3) The integration of sensing, communications, and control technologies with field devices in distribution systems to improve reliability and efficiency^[4].

In the meantime, renewable energy has come of age. The integration of renewable energy, especially wind and solar energy, reduces CO₂ gas emission and its impact on the environment. As wind and solar energy have developed, they are now accepted as viable options to traditional energy, such as oil and gas, for distributed and bulk generation^[5-6]. Consequently, their integration into existing grid has become increasingly important. Microgrids are a natural vehicle for solar and wind energy grid integration because of their local energy balancing and autonomous control nature of

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operation^[7-8]).

With the availability of energy balancing supported by storage technology and autonomous control, the concept of microgrid was extended to define a structured microgrid^[9]. Refer to Fig.1. A structured microgrid has the following definition: Integrated with loads, energy sources, storage devices, sensors, and data bus, a structured microgrid is an autonomous subsystem that features 1) Balanced energy over the intended operation/capacity; 2) Reconfigurable for standalone or grid connected; 3) Resilient with fault tolerance/fault isolation; 4) Bidirectional power flow; and 5) Modular/scalable.

Recently, grid resiliency is attracting more attention as weather events are becoming increasingly more frequent and damaging and as security threats external to the grid are increasing^[10-11]. Grid resiliency has three basic fundamental aspects: fault tolerance, fault isolation and fast electronic protection. Fault tolerance requires the circuit to be able to withstand a failure, such as a short, while maintaining critical performance.

Fault isolation demands the circuit to localize any failure to prevent cascading failures. Fast electronic protection reacts to protect hardware in a timely fashion or to reconfigure the circuit to minimize or prevent further damages to surrounding circuits.

A modern grid has 9 major areas as illustrated in Fig.2. They are: 1) Microgrids; 2) Physical grid; 3) All things grid connected (ATGC); 4) Command and control; 5) Sensor network/telemetry; 6) Communication network and security; 7) System integration and resiliency; 8) System analytics and big data; and 9) Policy, economics, and market. Power electronics and systems technology is playing an increasingly important role in the drive for grid modernization, since seven out of the nine of the major areas have power electronics content in it. Among all the desired characteristics for a modern grid, the electronicization is the very first step, since it lays the foundation for the modern and electronic grid. It is with this understanding that this article is written to bring more attention and more coordinated efforts together to move the grid modernization forward.

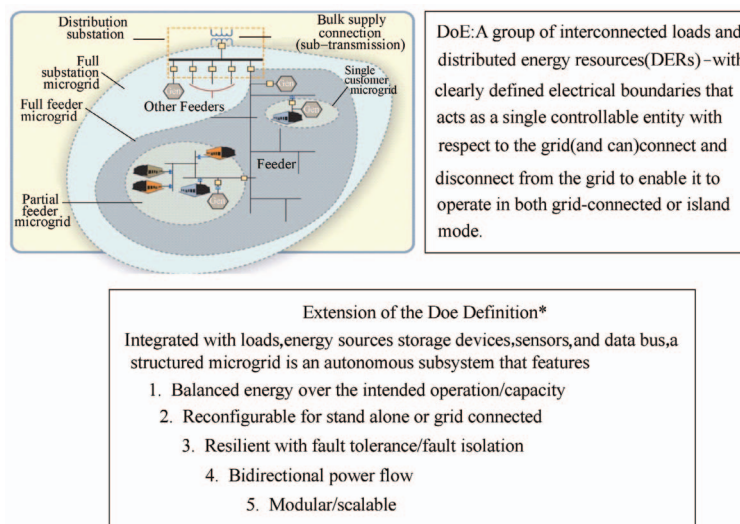


Fig.1 Structured microgrid: a definition

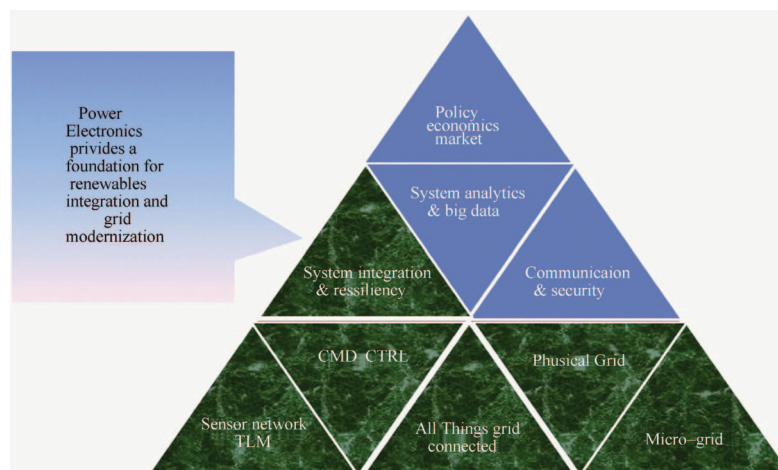


Fig.2 Power electronics: laying the foundation for a modern grid

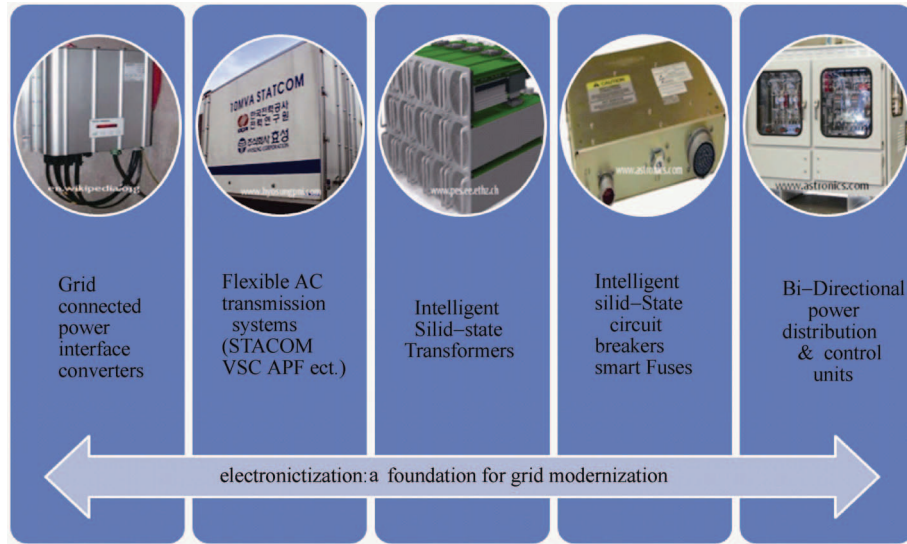


Fig.3 All things grid connected (ATGC): a foundation for grid modernization

There are 5 major categories of the hardware that is connected to an electronic power grid, as illustrated in Fig.3. They are: 1) Grid-tied and grid-supporting (smart) inverters; 2) FACTS supporting hardware; 3) Grid-scale electronic power transformer; 4) Solid-state circuit breakers, current limiters, smart fuses and sensors, and 5) Multi-terminal bidirectional power and control unit. For easy reference, it is frequently referred to as “All Things Grid Connected” (See, for instance[7], for more details).

This paper is organized into 6 sections. Section 1 is the introduction. Section 2 discusses grid-tied and grid-supporting inverters. Section 3 presents flexible AC transmission systems (FACTS) technology and hardware. Section 4 highlights the importance of the grid-scale electronic power transformer and its great potential. Section 5 introduces solid-state circuit breakers, current limiters, smart fuses and sensors. And Section 6 presents multi-terminal bidirectional power and control units that are essential for the retrofitting of power distribution substation. We finish the paper with concluding remarks and references.

2 Grid-tied and grid-supporting (smart) inverters

Recent interest in grid-tied converters started in the 1980's. Grid-tied functions typically include synchronizing with the grid with proper voltage and frequency, protection against unintended islanding mode of operation, and connect and disconnect capability (as Fig.4). The technology has since matured over the years, especially in the last decade, because of the proliferation of solar and wind energy for distributed generation and integration into the power grid. Many text books are now available (see [12], for instance).

Grid-tied inverters have to support a set of standards in order to be connected. They are, for instance, IEEE 1547 for interconnection of distributed

generation, IEC 61727 for characteristics of utility interface, VDE 0126-1-1 for safety, IEC 61000 for electromagnetic compatibility, and EN 50160 for public distribution voltage quality.

In recent years, research and development efforts have been focused on grid-supporting features of inverters, otherwise known as smart inverters. Smart inverters integrate the traditional required functions for inverters with grid supporting functions such as grid voltage regulation/var compensation, frequency adjustment, power flow, etc. Fault control is another important benefit that a smart inverter can bring to a grid. Fault ride through is a performance parameter that is used most often. Grid communication is also required for command, telemetry, and state-of-health reporting/monitoring^[13-19].

The challenges for smart inverter design include limited availability of energy to provide “inertia,” inadequate short circuit capability to clear a fault, lack of distributed control to coordinate multiple inverters under different conditions, negative dc incremental resistance introduced by active control for constant power output, and the upcoming multiple national and international regulatory requirements^[20].



Fig.4 Grid-connected (I/F) converters

It is anticipated that many of the challenges can be overcome by the introduction of storage and local autonomous control to form structured microgrids^[21-22]. A structured microgrid can maintain energy balance to a prescribed level and autonomously maintain resiliency for dead bus recovery or black start^[23-26].

3 FACTS-supporting hardware

Refer to Fig.5. The flexible AC transmission systems (FACTS) concept was first introduced in the 1980's, but the first piece of hardware was actually developed as early as in 1974 - a shut static var compensator (SVC) by Westinghouse. A generic definition of FACTS technology is: alternating current transmission systems incorporating power electronic-based and other static controllers to enhance controllability and to increase power transfer capability. Over the years, FACTS technology has been widely accepted. Typical hardware types include: Static Synchronous Compensators (STATCOM), Active Power Filter (APF), Unified Power Flow Controllers (UPFC), Convertible

Series Compensators (CSC), and many others^[27-29].

The benefits of FACTS hardware include more effective power flow control and increased transmission capacity. More specifically, FACTS offers: More effective control of power so that it flows on the prescribed transmission routes; More secured loading of transmission lines to levels nearer their thermal limits; Greater ability to transfer power between controlled areas; Virtually limiting cascading outages by isolating the effects of faults and equipment failure; and Damping of power system oscillations^[29].

The development and deployment of FACTS facilities has lagged behind expectations because it requires a relatively large facility foot-print, has a relatively higher cost, and is usually a location and system specific design.

4 Grid-scale (intelligent) electronic power transformers

Refer to Fig.6. The grid-scale electronic power

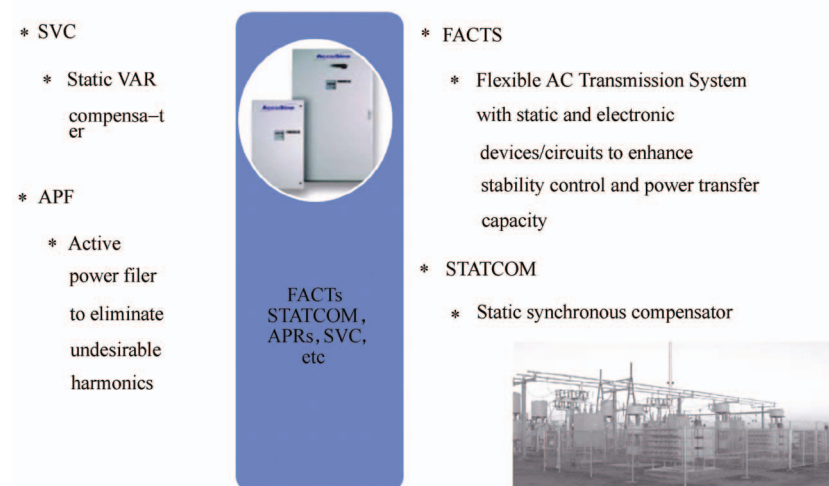


Fig.5 FACTS, STATCOM, APF, SVC, etc

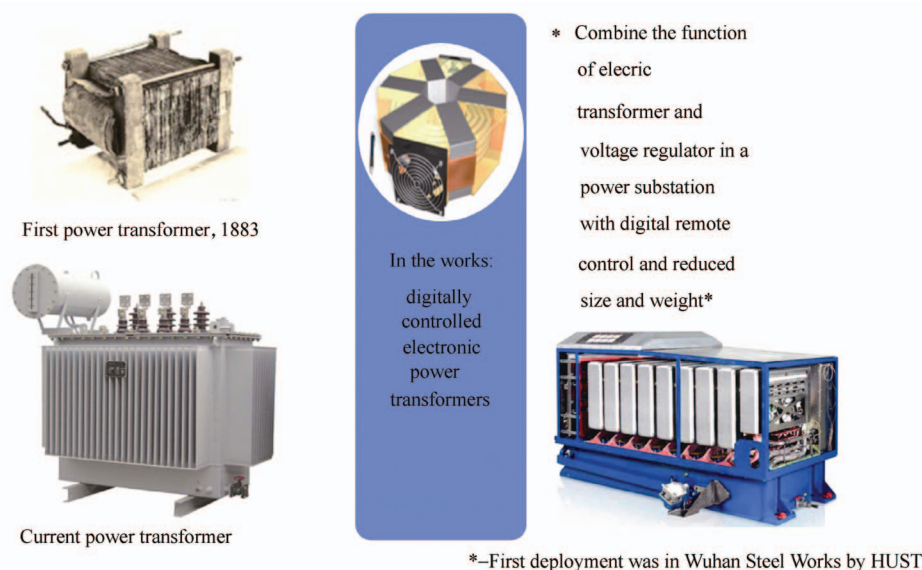


Fig.6 Grid-scale electronic power transformers

transformer development has attracted a lot of attention lately. It first appeared in literature as early as the 1970's^[30], but recent development has been more focused on grid and traction applications at high or medium high voltage levels ([31-35], for instance). Various names have been used, but we use grid-scale electronic power transformer to capture the essence that it still uses electromechanical structure for energy transferring, unlike a solid-state relay where the entire electromechanical relay is replaced with solid-state devices. An adjective “grid-scale” can be used to emphasize the fact it is focused on grid applications for bulk transmission and distribution.

The appeal for replacing the electromechanical power transformer with an electronic (and electromechanical) power transformer is twofold: to reduce the size/weight and to introduce more functionality with greater electronic control. The first function is obtained by use of power semiconductor devices to switch at a high frequency and the second function is introduced together with the controllability of power semiconductor devices. Various topologies and control methods have been proposed in the literature with successful prototypes built by industry and academic alike.

But the deployment of electronic power transformers has been lagging behind. There are a few major challenges in producing highly reliable and cost-effective electronic power transformers for grid applications. The first is the conversion efficiency. For traditional electromechanical power transformers, because of their simplicity, the overall conversion efficiency is typically around 98.5%. For switching converters this is a nontrivial challenge, as it requires the most advanced conversion technology known as adiabatic power conversion technology^[21]. In addition, careful design for circuit, electrical, electromechanical and thermal is required. The second challenge is the reliability of the transformer. Electronic power transformers are particularly vulnerable for over-voltage and over-current stresses when a large transient is induced, whereas a traditional power transformer is usually significantly more robust as long as the thermal stress is not exceeded. A third challenge is that with current state-of-the-art semiconductor power devices, the foot-print for an electronic power transformer is large at the present (similar to the case for FACTS technology). The fourth challenge is the anxiety typically associated with any new technology: people want to see proven technology before deployment for high reliability applications such as grid, but do not want to be the first one to use it.

However, new power devices, circuit topologies and control technologies are emerging to address these challenges and to replace traditional power transformers. New multi-phase, multi-level, and time-multiplexed power converters^[9], leveraging the latest component technologies and control methodologies, are improving efficiency and reliability. The wide-band-gap devices, such as GaN and SiC, are coming

of age, particularly SiC for high power applications (See, for example, [36]). With electronicization, a full-set of control capability can be added to an electronic power transformer for grid, similar to digital control for DC-DC power converters^[37]. With additional features and functionalities, the electronic transformers can offer added value to consumers and grid owners and operators. An electronic tap changer is one good example^[38]. New solid-state circuit protection devices are also emerging to enhance fault tolerance and fault isolation capabilities for grid-scale electronic power transformer with intelligence (More on this in Section 5).

5 Solid-state circuit breakers, current limiters, smart fuses, and sensors

Refer to Fig.7. Solid-state circuit breakers, current limiters, smart fuses and sensors are essential for fault-tolerant design, fault isolation and grid resiliency (including black start). Yet this is one area where the development and maturity has lagged behind many of the other areas we previously discussed. Therefore significant development is overdue and successful development will significantly improve grid autonomous operation, grid reliability and grid resiliency

Typical electromechanical circuit breakers react slowly to fault conditions, from hundreds of milliseconds to seconds. This is way too slow for a protection circuit, especially electronic power devices, to protect the circuit and equipment and to localize its over voltage and current surges. Typically, AC contactors take 3-6 cycles (at 50 or 60 Hz) to react and their price is low; circuit breakers take 3-6 cycles with low to medium price; and thyristor-based circuits take sub cycles with relatively high price. By contrast, solid-state electronic protection circuits can react in the order of microseconds and hence are essential for effective fault isolation to prevent any failure propagation. Circuit design needs to also consider failure modes to prevent failure propagation. Thorough failure modes and effects analysis (FMEA) are usually required.



Fig.7 Solid-state circuit breakers, current limiters, smart fuses, and sensors

Solid-state fault current limiters were discussed in depth in the early 1990's (See [39-41], for example). Recent efforts can be found in [41] through [44]. Recent overviews of the topics were present in [45] through [47]. From the technical domains, it can be classified as DC and AC types. From its mechanism, it can be classified into superconducting, purely solid-state devices, and electromagnetic. From their applications, it can be in three major areas: grid, microgrid and ship board.

Like any protection circuit, the location of the fault protection circuit is important due to its large foot print and relatively high cost. The technical challenges lie in the circuit robustness and ability to protect itself from unforeseen events that induce large transient in voltage, current, and energy (power). Another challenge to its limiting capability is system and location specific, similar to the case for FACTS technology.

Smart fuses are programmable and resettable fuses that are implemented with solid-state power devices and associated circuitry for control and self-protection. It provides connectivity by command after a fault is cleared or maintenance is completed. With programmability and resetability, smart fuses can also provide an effective fault management means for critical circuits and equipment when a redundancy is impractical because of prohibitive cost or size, weight, and power limitation. Smart fuses are effective in dealing with "smart shorts."

Sensors are important for a modern grid to reach its full capability for command and control. They will provide the basic state-of-health information and real-time data about the grid's physical parameters to monitor and to control. Basic physical parameters to be sensed include: voltage, current, power, watt hours, frequency, phase angle/power factor, harmonics, etc. With the proliferation of electronic technologies, smart sensors are becoming popular. Smart sensors incorporate post processing capability into the sensor before it is transmitted. They feature analog-to-digital converter, self-identification, real time clock, calibration, and data logging. Some may include a local microprocessor^[48-49].

Equipped with standard data bus, smart sensors can be networked together to provide unprecedented capabilities with two-way communication. Furthermore, a wireless sensor network (WSN) consists of distributed sensors to monitor physical parameters and environmental conditions, such as temperature, vibration, pressure, and to transmit data through the network to a grid operator's central office or a substation control room. One example is the wireless sensor networks for automated meter infrastructure (AMI). The added capability includes smart capacitor sensors to monitor and control capacitor banks remotely, for outage detection, for transformer monitoring, for high-voltage line temperature, and weather conditions^[50].

6 Multi-terminal bidirectional power & control units

Refer to Fig.8. Recently, the integration of solar

and wind energy to support grid functions is drawing a lot of attention. One of the approaches is to use the magnetic structure to couple multiple renewable energy sources together. A series connection of photo-voltaic panel and wind turbine generator was reported in the early 1990's^[51]. At the same time, a two-input flyback converter was reported in [52], where magnetic coupling is used to add energy together and the control was phase staggered (time multiplexed). A parallel connection approach was reported in [53], using a DC-bus to couple the multiple sources together. This approach is similar to a microgrid where the coupling is through various control of individual sources through a local controller.

Because of the limitation in maintaining the required voltage regulation, a magnetically coupled three-port approach was reported in [54], where multiple energy sources provide power together (in parallel) through magnetic coupling. A systematic approach was presented in [55], where a family of multi-port bidirectional DC-DC converters was derived and general characteristics discussed. The bidirectional power flow is an essential feature for grid applications and is accomplished through circuit symmetry in the circuit topology.

Many authors have published their three-port converters (See, for instance, [56] and [57]). It is worth noting that a particular magnetic couple approach was reported in [58], where a modular multi-level converter was used together with a medium frequency link. This approach leverages the advantages of modular multi-level converters to accomplish smaller size and weight. It is anticipated that multi-level converters will enable significant technological progress in multi-port converters. Recent new product offerings are a strong indicator (See, for instance, [59-61]). The synergy among grid-scale electronic power transformers, multi-port converters for renewable energy integration, and electric vehicle charging stations will propel the multi-port, multi-level and multi-phase technology to achieve significant progress in the near future to bring unprecedented performance and cost advantages to move grid modernization forward.

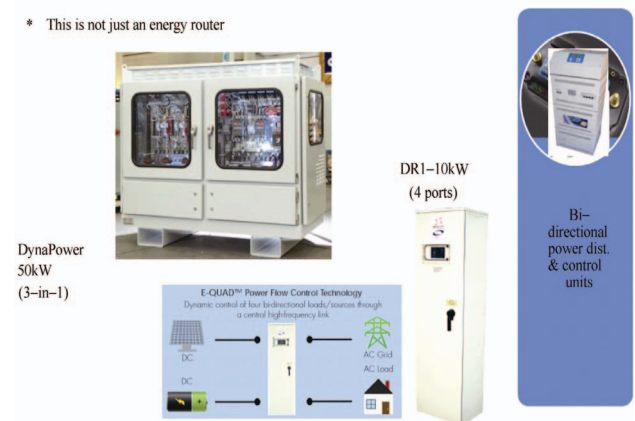


Fig.8 Bi-directional power & control unit

It is also anticipated that the multi-port technology can be potentially coupled with electronic power transformer technology to form a multi-port electronic power transformer that is essential for distribution substation modernization, together with renewable energy integration.

7 Conclusions

Electronicization is the first step towards a smart modern grid. It is the process that transforms the grid from the traditional electrical and electromechanical (EE) to electronic, electrical and electromechanical (EEE), therefore laying down the very basic foundation for a modern grid.

All things grid connected refers to 5 basic hardware groups that are needed for grid electronicization. They are: 1) Grid-tied and grid-supporting inverters; 2) FACTS-supporting hardware; 3) Grid-scale electronic power transformers; 4) Solid-state (electronic) circuit breakers, current limiters, smart fuses and sensors; and 5) Multi-port bidirectional power & control units.

Development and deployment of ATGC are two fundamental challenges for grid modernization. Grid modernization is a grassroots drive to transform the grid from the old passive technology to new active technology through electronic power transmission, distribution, processing and protection. Grid modernization also represents a win-win-win situation for the environment, consumers and grid owners/operators.

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